TOPIC 2 Systems of Linear Equations and Inequalities

Name _____ Date _____

Reminders
System of inequalities: graph and shade each one; the overlap is the solution set.
A solution makes EVERY inequality true. Points on a dashed line are not solutions.
In a story: no negative amounts, so stay in the first quadrant.

1. Is (1, 1) a solution of $y > 2x - 3$?

Tip: substitute the point

A. No, because $1 < -1$

B. No, because $1 > 2$

C. Yes, because $1 > -1$

2. Which inequality has a solid boundary line with the half-plane above shaded?

Tip: solid → ≥ or ≤

F. $y \geq x + 1$

G. $y > x + 1$

H. $y \leq x + 1$

3. Which point is a solution of $x + y \leq 6$?

> **Tip:** add, then compare

A. (4, 3)

B. (2, 3)

C. (6, 1)

4. Which inequality means “at most 10 hours”?

> **Tip:** at most → ≤

F. $h \geq 10$

G. $h < 10$

H. $h \leq 10$